

FMEA - CampusRide v4.4 subsystem failure-mode analysis

RPN = Severity x Occurrence x Detection. Anchored to draft.md §5.10 deployment snapshot, §5.16 HoQ, §6 audit, §7.2 limits.

| #  | Subsystem                | Failure Mode                                       | Failure Effects                                | Possible Cause                                   | Detection Conditions                           | S | O | D | RPN | Risk Level         |
|----|--------------------------|----------------------------------------------------|------------------------------------------------|--------------------------------------------------|------------------------------------------------|---|---|---|-----|--------------------|
| 1  | Auth / Resend Email      | Verification email not delivered                   | Users stuck in pending state; cannot log in    | Resend rate-limit or email flagged as spam       | Login-failure rate spike; Resend dashboard 4xx | 4 | 2 | 2 | 16  | Medium-High        |
| 2  | Ride-Match               | Race on last seat -> double-booking                | Two riders confirmed for same seat; conflict   | Concurrent POST without row-level lock on rides  | seats_remaining < 0 in ride table logs         | 4 | 2 | 2 | 16  | Medium-High        |
| 3  | Cost-Splitting / Payment | Payment gateway not integrated                     | Manual settlement only; no escrow protection   | Design-only in v4.4; Stripe not wired            | N/A current; design risk documented in §7.2    | 3 | 4 | 4 | 48  | High (design risk) |
| 4  | SOS / Safety             | SOS button not wired to emergency contacts         | Rider in distress lacks automated escalation   | Design-only stub; no backend service             | Zero invocations to date; feature flag off     | 5 | 3 | 4 | 60  | High               |
| 5  | Group-Chat Auto-create   | Duplicate-key 23505 on first booking insert        | Brief 500 to client; chat eventually OK        | Postgres unique constraint on (ride_id, user_id) | Backend log shows code 23505 swallowed         | 1 | 2 | 2 | 4   | Low                |
| 6  | Notification Bundle      | 6-notification fan-out fails partway               | Driver missing booking notice; rider notified  | Partial sendNotification failure; no transaction | notifications.unread per ride mismatched       | 3 | 2 | 2 | 12  | Medium             |
| 7  | Rating System            | Update-in-place allows downward revision           | Driver retaliation risk; F5-validated concern  | No dispute window; no rating freeze logic        | F5 driver subset N=19 signals rating-anxiety   | 4 | 3 | 2 | 24  | Medium-High        |
| 8  | Points Subsystem         | point_rules empty; point_transactions missing      | Points always 0; motivation gap                | Migration not run on Railway production DB       | REST returns PGRST205 on point_transactions    | 3 | 5 | 1 | 15  | Medium             |
| 9  | WeChat Outreach          | Mini-program short-link expires; H5 fallback fails | Outbound push silently broken; no WeChat reach | wechat-link.service exception not retried        | wxgroup_notice_record dispatch_status='failed' | 3 | 2 | 2 | 12  | Medium             |
| 10 | .edu Whitelist           | Single-tenant: only @cornell.edu accepted          | Multi-campus expansion blocked; EC12 pending   | Institution-domain whitelist hard-coded          | Manual test fails for non-Cornell .edu addr    | 3 | 4 | 1 | 12  | Medium             |
| 11 | Geo-Checkin              | User fakes GPS to claim activity points            | Inflated points balance; fraudulent rewards    | No device attestation; client-trusted geo        | device_info JSONB inspection (manual)          | 2 | 2 | 3 | 12  | Low                |
| 12 | Group Discovery          | All 5 community groups have member_count=1         | No flywheel; community module cold-start       | Discovery UI not yet surfaced in app shell       | Manual review of groups; DB count check        | 3 | 4 | 2 | 24  | Medium             |

Scale (S, O, D):

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|



RPN gradient (1 -> 80): 



Risk: Low Medium Medium-High High

